When Graphs Meet LLMs: Graph Learning Enhancement and Scientific Applications

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Abstract. This dissertation presents a unified framework for integrating graph structures with large language models (LLMs) to combine relational reasoning and semantic understanding. First, a decoupled structural aggregation method selectively incorporates semantically and structurally relevant neighbors as inputs into LLMs, enabling scalable and effective node classification. Second, the paradigm named graph-as-new-language reformulates graph topology as node-to-center paths, allowing LLMs to internalize structural patterns directly through sequence modeling and task-specific finetuning. Third, scientific applications validate the framework by integrating graph with language models in medical question answering and drug discovery domain. A graph-based module detects missing clinical information via subgraph comparison for self-verification enable LLMs understand and give responses correctly. Moreover, a multi-objective framework based on LLMs generates toxicity-guided molecular analogues while preserving scaffold structures and satisfying structural constraints in structure based drug discovery. Collectively, these contributions address challenges in structural–sequential mismatch, input-length limitations, and joint reasoning, demonstrating scalable, interpretable, and application-driven graph modeling.

Keywords: Large language models, graph learning, structural aggregation, graph language, scientific applications

1 Background and Motivation

Graph Learning holds significant importance in both practical applications and theoretical research. In terms of real-world applications, graph structures serve as a powerful tool for modeling complex relationships. Key application domains include social network analysis, recommender systems, financial fraud detection in transaction networks, medical question answering, and drug discovery. In these scenarios, entities and their interactions naturally form networks, providing a direct and applicable arena for graph learning methods.

Regarding the research necessity of graph structure, the graph paradigm offers unique and indispensable advantages: it natively and efficiently models the connections between nodes, allowing for the explicit representation of a system's internal structure, dependencies, and complex interactions. This explicit encoding of relationships forms the foundation for achieving reliable understanding and enabling multi-hop reasoning within complex systems. But this format is not directly designed for large language models with input of sequences [2].

Facing GNNs' weaknesses in handling text and the limitations of simple cascade models [3], the field is increasingly turning to LLMs for their strong understanding, generalization, and open-world reasoning. We argue that combining these semantic strengths with explicit graph relational representations can overcome prior limitations to enable richer relational understanding and reasoning.

2 Challenges and Research Questions

Recent advances in large language models (LLMs) have demonstrated remarkable capability in semantic understanding and reasoning, while graph learning provides an expressive paradigm for modeling relational structures in complex systems such as medical question answering. However, these two paradigms remain largely disjoint in their representational assumptions and learning mechanisms. This dissertation addresses the fundamental question of how to unify structured relational reasoning with language-based intelligence.

Specifically, this research is guided by the following questions:

1. How can graph structural information be incorporated into LLMs in a way that preserves relational fidelity?
2. Can graph topology be reformulated as a language-compatible representation that enables LLMs to internalize structural patterns?
3. To what extent can unified graph-language modeling enhance reasoning and overall task effectiveness in scientific applications, particularly medical question answering and drug discovery?

3 Related Work

LLMs have already been widely applied to graph learning, but current methods cannot achieve a deep integration of the LLMs' generative capabilities with graph structure. Existing methods incorporate structural information either by encoding it as natural language prompts, such as SimCSE [4], or by augmenting node texts with neighborhood information, such as WalkLM [5] and InstructGLM [7]. These approaches suffer from information loss and input-length limitations. More recent frameworks, including GraphGPT [6] and LLAGA [1], align graph embeddings with textual representations to reduce redundancy, but they rely primarily on local structures and achieve limited structural–semantic integration. Few of these methods could preserve the relations that Graph Neural Networks use, so LLMs cannot process them.

4 Research Plan and Contributions

This dissertation presents a unified framework for integrating graph learning with language models in three directions. The main contributions are as follows:

1. **Structural aggregation for graph learning.** A decoupled aggregation method combines semantic and structural context by selecting informative neighbors instead of entire neighborhoods, improving scalability under sequence-length constraints and enhancing node classification.
2. **Graph-as-language modeling.** Graph topology is represented as language-like node-to-center paths, enabling LLMs to capture structural patterns through sequence modeling and task-specific adaptation by concatenating related paths.
3. **Scientific applications.** In medical question answering, a graph-based module identifies missing clinical information of user questions for self-verification. Furthermore, in structure-based drug discovery considering molecules as graphs, a multi-objective framework based on LLMs generates toxicity-aware analogues under structural constraints for interpretable molecular design.

5 Timeline

Phase 1: Structural Aggregation for LLM-based Graph Learning (Completed / Published Work)
Phase 2: Graph-as-Language Modeling Framework (Completed / Under Review)
Phase 3: Scientific Application Validation (Ongoing → Planned Completion)

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